

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination 2022

ESE 3211

(Coal Power Generation)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|--|----|
| 1(a). | Discuss about the different types of coal including their applications. | 12 |
| 1(b). | Why are proximate and ultimate analysis of coal necessary? Explain. | 08 |
| 1(c). | What are coal preparation and coal storage? | 05 |
| 1(d). | Exactly 2.5 g coal was weighted into a crucible. After heating for 1 hour at 110°C the residue weighted 2.415 g, the crucible was then covered with a vented lid and strongly heated for 7 minutes at $900 \pm 15^\circ\text{C}$. The residue then weighted 1.528 g. The crucible was then heated without cover at 700°C until a constant weight was obtained. The last residue was found to weight 0.245 g. Calculate the percentage result of above analysis. | 10 |
| 2(a). | What is coal handling? Differentiate between in-plant and out-plant coal handling approaches. | 06 |
| 2(b). | What is coal recovery? Which factors should be considered while selecting a coal mining method? | 09 |
| 2(c). | What are the different coal mining methods? With the help of necessary diagram, describe the slope mining of coal. | 10 |
| 2(d). | Which operations are carried out for cleaning the coal? Describe the dense media washing method with the help of appropriate figure. | 10 |
| 3(a). | Write down the coal preparation steps. How quality of coal can be improved by Jig-table washing? Discuss. | 10 |
| 3(b). | What are the different types of industrial dust collectors? With the help of a schematic diagram, describe the working principle of centrifugal dust collector. | 12 |
| 3(c). | Briefly describe the principle upon which a dust collection system works. | 08 |
| 3(d). | Discuss the benefit of fly ash. | 05 |
| 4(a). | Define coal firing and coal devolatilization. Differentiate between over-feed and under-feed stokers. | 08 |
| 4(b). | Why coal is pulverized? Describe the central system for burning pulverized coal. | 15 |
| 4(c). | With the help of necessary sketch, explain the cyclone burner. | 12 |

SECTION – B

- | | | |
|-------|---|----|
| 5(a). | What is meant by clean coal technology? Why it is necessary? | 05 |
| 5(b). | Define CO ₂ sequestration? Explain the Integrated Gasification Combined Cycle plant which included with CO ₂ capture. | 10 |
| 5(c). | With necessary sketch, explain a chemical process for CO ₂ capture from exhaust flue gas. | 10 |
| 5(d). | Why captured CO ₂ is needed to store? Explain ocean storage system of CO ₂ . | 10 |
| | | |
| 6(a). | Why the emission of SO _x and NO _x should be regulated? Briefly explain. | 08 |
| 6(b). | What is fluidized bed combustion? Why it is called fluidized bed combustion? Mention its advantages. | 10 |
| 6(c). | Upon which criteria steam power plants are classified as subcritical, supercritical and ultra-supercritical? What are the reasons to develop ultra-supercritical steam power plant? | 10 |
| 6(d). | What is boiler specification? Why boiler mountings and accessories are used in boiler? | 07 |
| | | |
| 7(a). | What is boiler? Why is it used? | 06 |
| 7(b). | What are the limitations of La Mount boiler and Benson boiler? How these limitations can be overcome by Loeffler boiler? | 09 |
| 7(c). | Differentiate between boiler mountings and accessories with appropriate example. | 08 |
| 7(d). | In a steam power plant, steam generation rate is 4800 kg/h, fuel consumption rate is 450 kg/h. The boiler generating steam at a pressure of 16 bar and temperature of 305°C. Temperature of water entering the economizer at 27°C and temperature of water leaving the economizer at 90°C. If the calorific value of fuel is 28 MJ/kg, determine the followings:
(i) Equivalent evaporation rate, (ii) thermal efficiency of boiler, and (iii) percentage of heat utilized in boiler and economizer. | 12 |
| | | |
| 8(a). | Define turbine and steam turbine. Why nozzles are used in steam turbine? | 07 |
| 8(b). | Write down the differences between impulse and reaction turbine. | 06 |
| 8(c). | Why should steam turbine be governed? With necessary diagram, explain the nozzle control governing of steam turbine. | 10 |
| 8(d). | Why is compounding of steam turbine necessary? In a single stage impulse turbine, the nozzle angle is 30° and blade speed is 400 m/s. The steam issues from the nozzle to the turbine with a speed of 750 m/s. The blades are symmetrical. Assume blade friction factor is 0.87 and $\eta_{nozzle} = 92\%$. For a steam flow rate of 850 kg/h, calculate the (i) blade angles, (ii) work done per kg of steam, (iii) power developed, (iv) blade efficiency, (v) stage efficiency. | 12 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2022

ESE 3203

(Petroleum and Natural gas processing)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | Briefly describe crude oil compositions. What are the impurities that are associated with a crude oil? | 10 |
| 1(b). | What is the octane number, and how is it used to evaluate the anti-knock performance of gasoline? | 07 |
| 1(c). | Explain briefly why crude oil needs to be refined and how this is done? | 07 |
| 1(d). | The specific gravity of NGC is 0.802, determine the API gravity of NGC. | 05 |
| 1(e). | Define : i) Octane number, ii) API gravity, iii) Flash point | 06 |
| | | |
| 2(a). | Describe the crude oil desalting process and accompany it with a necessary diagram. | 10 |
| 2(b). | Explain atmospheric distillation and vacuum distillation of crude oil showing the arrangement of towers to obtain principle fractions. | 10 |
| 2(c). | What is catalytic reforming? Briefly describe the major reactions in a catalytic reforming process. | 08 |
| 2(d). | What do you understand by True boiling point distillation (TBP)? Explain. | 07 |
| | | |
| 3(a). | Discuss ASTM distillation characteristics with a neat diagram. | 08 |
| 3(b). | Describe the delayed coking process and provide a neat sketch to illustrate the key steps. | 10 |
| 3(c). | What is visbreaking? Differentiate between coil visbreaker and soaker visbreaker. | 07 |
| 3(d). | With the help of a neat labeled diagram explain the working of a fluid catalytic cracking (FCC) unit. | 10 |
| | | |
| 4(a). | What are the objectives of product blending? Describe how to calculate RVP blending in the product. | 08 |
| 4(b). | Give a detailed description of "lube oil" processing with an appropriate diagram. | 08 |
| 4(c). | How one can estimate the viscosity of a product blend? Estimate the viscosity of a blend of 2000 barrels of Fraction-1 with viscosity of 72 cSt, 3000 barrels of Fraction-2 with a viscosity of 100 cSt, and 4000 barrels Fraction-3 with a viscosity of 201 cSt, all at 130°F. | 10 |
| 4(d). | With the help of a neat labeled diagram, explain the hydrotreating process. | 09 |

SECTION – B

- 5(a). What is natural gas condensate? With the help of a flow diagram, show the separation of condensate from raw natural gas. 12
- 5(b). Describe the most prevalent theory of natural gas formation with a neat sketch. 13
- 5(c). What are the differences between porosity and permeability? A sandstone has a void volume of 35 cm^3 and the total volume of the rock is 83 cm^3 , measure the porosity. 10
- 6(a). Demonstrate the design and operation of a 'Vertical Separation' with an appropriate figure. 15
- 6(b). What are the functions of, (i) Centrifugal Inlet diverter, and (ii) Ware Breakers. 10
- 6(c). An incompressible fluid flows in a linear process media
(where, $q = 0.001127 \text{ kA} (P_1 - P_2) / \mu L$ with the following properties: $L = 2000 \text{ ft}$, $h = 20 \text{ ft}$
width = 300 ft , $k = 100 \text{ md}$, $Q = 15\%$, $\mu = 2 \text{ cp}$, $p_1 = 2000 \text{ psi}$, and $p_2 = 1990 \text{ psi}$)
Calculate, 10
- a). Flow rate in bbl/day
- b). Apparent fluid velocity in ft/day
- c). Actual fluid velocity in ft/day.
- 7(a). Discuss Alkaline Amine Process for gas sweetening with an appropriate block diagram. 12
- 7(b). Why is dehydration process a must in natural gas processing? 08
- 7(c). What is LNG? Describe its properties with a schematic of a LNG plant, describe the overall LNG production process. 15
- 8(a). Explain the working principle of an orifice meter. 10
- 8(b). State the potential hazards and remedies occurred from LPG. 15
- 8(c). Find the water content of a gas at 1000 psia and 100°F using compbells method. The gas composition is as follows : $\text{CH}_4 = 80.0\%$, $\text{C}_2\text{H}_6 = 5.0\%$, $\text{C}_3\text{H}_8 = 1.5\%$, $n\text{-C}_4\text{H}_{10} = 0.5\%$, $\text{CO}_2 = 2.5\%$, $\text{N}_2 = 2.0\%$ and $\text{H}_2\text{S} = 8.5\%$ (also see the graph) 10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2022

ESE 3221

(Energy Storage Systems)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | How does the power of an energy storage system influence its application? Explain with diagram. | 10 |
| 1(b). | Why the axes of a Ragone chart are logarithmic? Explain. | 10 |
| 1(c). | Why is energy storage system is so important for renewables integration with grid. | 08 |
| 1(d). | Write down the equations of head for generating mode and pumping mode of pumped hydro storage system. Also mention what each term in the equation mean. | 07 |
| | | |
| 2(a). | How does compressed air energy storage work? Explain with diagram. | 10 |
| 2(b). | What is significance of high power density and high cycle life in the context of a flywheel energy storage system? | 06 |
| 2(c). | Why electrical machine is used in flywheel storage system? Draw a thematic diagram of flywheel storage system and show its components. | 12 |
| 2(d). | Between the pumed-hydro system and CAES system, which one will you choose and under what circumstances? | 07 |
| | | |
| 3(a). | Wht is the thermocline zone and what role does it play in storage tank? Explain with necessary figures. | 09 |
| 3(b). | Discuss the role of PCM in thermal energy storage. | 07 |
| 3(c). | How does the Aquifier thermal energy storage system provide heating and cooling for buildings? Explain with diagram. | 13 |
| 3(d). | How does the power conditionning system in a SMES unit function? | 06 |
| | | |
| 4(a). | How does an EDLC store and discharge energy? Describe with necessary figures. | 07 |
| 4(b). | How do the battery charges in off-line and line interactive systems difffer from those in a online double conversion UPS system? | 10 |
| 4(c). | Why are hybrid power plants becoming more prominent? Draw a hybrid power plant. | 11 |
| 4(d). | What is the role of a high power reversible AC–DC converter in providing system-level services to the grid in fast charging stations? | 07 |

SECTION – B

- 5(a). Define battery. What is the basic difference between a cell and a battery? 05
- 5(b). Why grids of a lead acid battery is made of alloys? 06
- 5(c). Explain how a VRLA battery works. What are the electrolyte immobilization techniques of a VRLA battery and how do these techniques facilitate the passage of oxygen to the negative plate. 12
- 5(d). Draw flow diagram of lead-acid manufacturing process with brief explanation. 12
- 6(a). Why is the practical energy output of a battery only fraction of theoretical energy? 06
- 6(b). Explain reaction mechanism of a Li-ion battery with necessary diagram. 11
- 6(c). What is solid electrolyte inter phase (SEI) layer? Mention briefly the root cause of degradation and various degradation modes in a battery. 10
- 6(d). Briefly describe various aging mechanisms of negative electrode of a Li-ion battery with thematic diagram. 08
- 7(a). Describe working principle of a VRFB with necessary diagram. Why cycle life of VRFB is so high? 11
- 7(b). Classify the modeling of battery? 06
- 7(c). New Tesla cyber truck comes with a better battery management system (BMS). Describe BMS architecture and key features. 10
- 7(d). If Tesla have vast amount of data about various parameters of their battery pack and they want to implement a new battery model that is more accurate than previous model (as Li-ion battery system is highly non-linear and conventional models are not fully accurate), then which type of battery modelling method should they use and why? 08
- 8(a). When is used underground hydrogen storage? 08
- 8(b). Hydrogen fuel cell can be used in modern EVs. Toyota Mirai consists of type IV compressed hydrogen storage tanks as fuel storage for fuel cell operation. Describe construction of type IV pressure vessel. 12
- 8(c). “Cryo-compressed hydrogen storage minimizes the challenges of both compressed and cryogenic hydrogen storage”. Explain. 08
- 8(d). What is H_2 economy? What is role of H_2 storage in H_2 economy? 07

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2022

ESE 3207

(Solar Photovoltaic systems)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | Define Fermi level in a semiconductor. Briefly explain the physical significance of Poisson and continuity equation together for semiconductor material. | 11 |
| 1(b). | Briefly explain the role of carrier generation and recombination in maintaining dynamic equilibrium. Define minority carrier diffusion length with physical significance. | 11 |
| 1(c). | Discuss in details about the balance limit of solar cell efficiency according to Schokeley –Queisser limit. | 13 |
| | | |
| 2(a). | What are the key characteristics of quartzite that make it a suitable source material for silicon wafer production? Discuss the production of silicon wafer from quartzite. | 13 |
| 2(b). | What are the various types of materials used in the construction of thin-film solar cells and how do they contribute to the overall efficiency of the cell? | 07 |
| 2(c). | What is lattice mismatch? Justify the use of Ge, GaAs and GaInP to form triple junction cells. | 08 |
| 2(d). | Why is a supporting layers are used in solar cells and what consideration are taken into account when selecting for this crucial component? | 07 |
| | | |
| 3(a). | Explain the working principle of a p-n junction solar cell using schematic diagram. | 10 |
| 3(b). | Draw and briefly explain the equivalent circuit of a realistic solar cell which considers all the losses, and also write the modeled equation. | 08 |
| 3(c). | Briefly explain the design rules for a solar cell with respect to loss. | 09 |
| 3(d). | A crystalline silicon wafer is doped with 10^{16} acceptor atoms per cubic centimeter. A 1 micrometer thick emitter layer is formed at the surface of the wafer with a uniform concentration of 10^{18} donors per cubic centimeter. Assume a step p-n junction and that all doping atoms are ionized. The intrinsic carrier concentration in silicon at 300K is $1.5 \times 10^{10} \text{ cm}^{-3}$. Calculate built in voltage, width of the depletion region and maximum electric field at the metallurgical junction. | 08 |
| | | |
| 4(a). | Why TCO is used in solar cell? Also explain why III-V material is considered for thin film solar cell technology. | 10 |
| 4(b). | Explain the operation of CdTe solar cell using layer structure. | 10 |
| 4(c). | Explain the challenges for multifunction ^{multijunction} solar cell. | 07 |
| 4(d). | What are the distinguishing features of perovskite solar cell from traditional silicon-based in terms of their construction and materials? | 08 |

SECTION – B

5(a).	What are the key components of a solar home system and explain the role of each component in the process of converting sunlight into usable electric energy.	10
5(b).	How does the surface area of a solar cell impact its output current, and what key factors should be considered in optimizing the design of a solar cell for efficient energy conversion?	10
5(c).	The maximum rated power point of a crystalline solar cell is 75Wp. Calculate the maximum power output at 45°C, 55°C and 65°C cell operating temperature. For crystalline silicon cell, decrease in maximum power per degree centigrade, increase in cell temperature (from STC value of 25°C) is $-0.45\% / ^\circ\text{C}$	07
5(d).	Why energy payback time is important of a PV system design? What factors influence the energy payback time of a PV system?	08
6(a).	How does mismatch in current affect the performance of series connected PV modules, and what strategies or technology are employed to mitigate current mismatch issues in order to optimize the overall efficiency of a solar PV system?	10
6(b).	What role does a charge controller play in solar system? How does incremental conductance algorithm function in MPPT charge controllers and what advantages does it offer over simpler algorithms like perturb and observe?	12
6(c).	A power converter has specified input power of 300 watts and manufacturer specified efficiency is 88%. How much is power loss within the converter? What is the output power?	07
6(d).	Design a solar PV system for a house which contains 3 fans of 70 watts each running 4 hours a day, 3 tubes lights of 35 watts each running for 8 hours a day and a refrigerator of 250 watts running for 12 hours a day, where consider compressor run 6 hours and off 6 hours. Consider battery autonomy 2 days, DOD 50%, battery charging efficiency is 80%, peak sun hours daily is 6, battery voltage is 12V.	06
7(a).	Discuss the working principle of on-grid solar PV system with storage.	12
7(b).	What is Net metering? How does it work?	10
7(c).	Discuss the effect of PV module temperature, orientation and tilt angle on the performance of a solar PV system.	13
8(a).	What are the basic parameters of a DC to DC converter? Write the basic working principles of DC-DC conversion of solar system.	10
8(b).	In what ways does temperature affect solar cell efficiency? What strategies and technologies can be employed to reduce the temperature related challenges in solar PV systems?	10
8(c).	Does solar panel reduce carbon footprint and how does production of solar energy contribute to carbon footprint?	07
8(d).	Can solar modules be reused after reaching the end of their life cycle, and if so, how? What environmental impact is associated with improper disposal of solar modules?	08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2022

ESE 3217

(Instrumentation and Control)

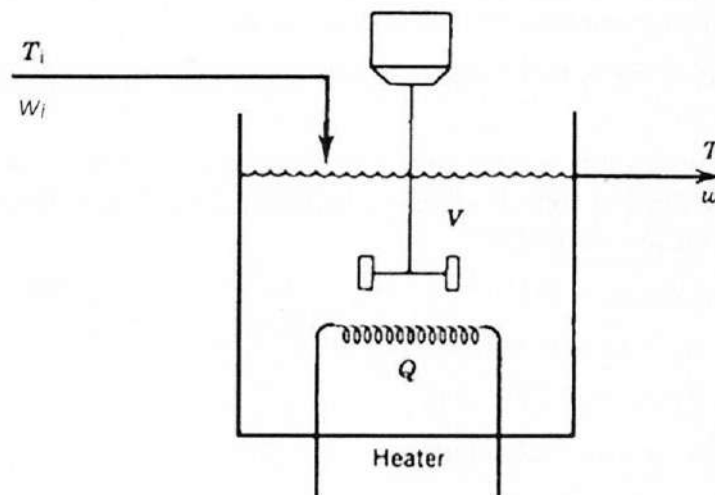
Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
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SECTION – A

- 1(a). Define process and system with example. Also explain different types of process variable with practical example. 10
- 1(b). Describe the general approaches to design a control system with appropriate example. 11
- 1(c). Consider a stirred-tank heating system shown in figure. The liquid inlet stream consists of a single component with a mass flow rate W_i and inlet temperature T_i . The tank contents are agitated and heated using an electrical heater that provides a heating rate of Q . Develop a dynamic model that describe the temperature of the component. Also, draw the control volume of the system. 14



- 2(a). Why is the use of Laplace transforms essential in the context of dynamic system and control engineering? Briefly classify the differential equation. 12
- 2(b). Solve the differential equation using Laplace transform. 15

$$\frac{d^3 y}{dt^3} + 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} + 6y = 1$$

with initial conditions $y(0) = y'(0) = y''(0) = 0$ where the primes denote derivatives.

- 2(c). Define transfer function with proper example. Also develop transfer function for a linear n^{th} -order ODE assuming all the initial conditions are zero. 08
- 3(a). With proper mathematical model, explain dynamic response of 1st order and 2nd order process. 10
- 3(b). Briefly explain the basic control system elements using block diagram. 11
- 3(c). Consider the third order linear time invariant system described by the following differential equation: 14

$$\frac{d^3 y}{dt^3} + 2 \frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 4y = 2u$$

- i. Define three state variable x_1, x_2 and x_3 based on the derivatives of the output y .
ii. Express the state equation in the matrix form.
iii. Write the output equation y in terms of the state variable.

4(a).	Explain fuzzy logic controller with block diagram.	08
4(b).	Explain cascade control system with block diagram. Between cascade and fuzzy control, which one will you choose and why? Discuss.	11
4(c).	Develop a block diagram illustrating the components of a parallel form of PID controller in the Laplace domain.	10
4(d).	Why differential equation is so essential in control system study?	06

SECTION – B

5(a).	What is meant by Piping and Instrument Diagram (P&ID)? Briefly explain the functional elements of a measurement system with block diagram.	13
5(b).	What is automation system? “Integrated automation system is the system that can ensure less human interaction”. Explain.	08
5(c).	Write short notes on signal conditioning element.	08
5(d).	Write short notes on (i) control valve sizing, (ii) sensor, and (iii) actuator.	06

6(a).	Describe parts of a control valve with necessary diagram. What is a quick opening valve and in which operation, it is appropriate to use?	10
6(b).	Why internal relay is important in a PLC system? Explain different types of timers in a PLC system.	12
6(c).	Water is pumped from one tank to another through a piping system with a total pressure drop of 150 psi. A control valve is provided in the piping loop to modulate the flow rates. Assume the key variables:	13

Fluid temperature = 70 °F

Design (maximum) flow rate = 160 gpm

Operating flow rate = 110 gpm

Minimum flow rate = 25 gpm

Pipe diameter = 3 inches

Specific gravity = 1.0

Valve characteristics = globe valve, equal percentage

Now size the control valve based on the chart below:

Size	C _v @10% VT	C _v @25% VT	C _v @50% VT	C _v @75% VT	C _v @100% VT
1 $\frac{1}{2}$ ''	0.83	1.88	7.5	12.5	25
2''	1.33	3.00	12.0	20.0	40
2 $\frac{1}{2}$ ''	2.10	4.73	18.9	31.5	63
3''	3.33	7.5	30.0	50.0	100
4''	5.33	12.00	48.0	80.0	160

7(a).	Explore the types of counters available in PLC. How do they differ in functionality?	11
7(b).	Why remote I/O module is advantageous?	06
7(c).	Draw the wiring diagram of PLC connection of a motor on-off control with signal lamp. The system is operated with on-push and off-push buttons. Two signal lamp is used to identify if the motor is on or off.	10
7(d).	Define relay. What are the differences between microprocessor and microcontroller?	08

- 8(a). Define the term protocol as it applies to a network. Explore the role and significance of human machine interface (HMI) in industrial automation. 08
- 8(b). What are the common types of wires used in data communication of a control system? "Parallel communication is more expensive than serial communication system". Explain the statement. 07
- 8(c). Briefly explain the working principle of DCS. Also write the difference between DCS and SCADA. 10
- 8(d). What is cascading timer? Use timers to do the PLC programming of a car wash system that consist of the following operations: 10
- When a sensor detects a car, a water spray system is activated instantly, and it is on for 60 seconds.
- Then the spray stops, and a nozzle is activated to spray soap water for 15 seconds.
- After that a robotic arm brushes the car for 120 seconds.
- Then again, the water spray is on for 120 seconds.
- After the final wash, a horn buzzes to indicate washing is done.

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